Sigma Notation

Exam code: 9MA0

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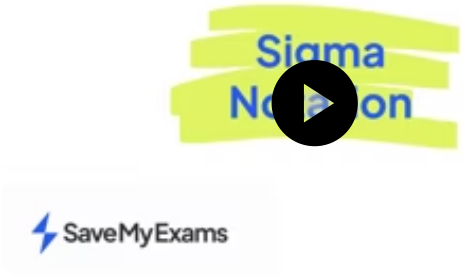
Updated on 20 June 2025

Help me understand



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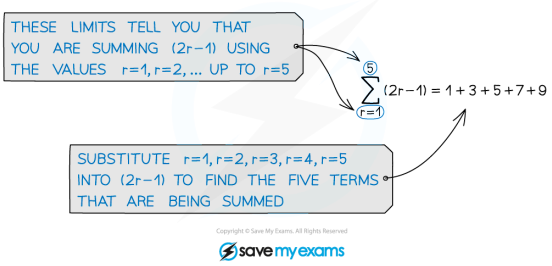
Sigma notation

What is sigma notation?

- The symbol Σ is the capital Greek letter sigma – that's why it's called 'sigma notation'!
- 'Σ' stands for 'sum' – the expression to the right of the Σ tells you what is being summed, and the limits above and below tell you which terms you are summing



- Be careful – the limits don't have to start with 1!
  - For example: $\sum_{r=0}^{4}(2r+1)$ or $\sum_{r=7}^{11}(2r-13)$

How do I use sigma notation?

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- **Geometric** will have the form $A \times B^r$
- Writing out the first few terms will help you

$$\sum_{r=1}^{5}(5r+1) = 6 + 11 + 16 + 21 + 26$$

THIS IS THE SUM OF THE FIRST FIVE TERMS OF THE ARITHMETIC SERIES WITH FIRST TERM 6 AND COMMON DIFFERENCE 5

$$\sum_{r=1}^{5}(4 \times 3^r) = 12 + 36 + 108 + 324 + 972$$

THIS IS THE SUM OF THE FIRST FIVE TERMS OF THE GEOMETRIC SERIES WITH FIRST TERM 12 AND COMMON RATIO 3

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- To work out such a sum use the arithmetic and geometric series formulae
- As long as the expressions being summed are the same you can add and subtract in sigma notation
  - For example:

$$\sum_{r=1}^{6}(4r+7) + \sum_{r=7}^{11}(4r+7) = \sum_{r=1}^{11}(4r+7)$$

$$\sum_{r=1}^{100}(7 \times 2^r) - \sum_{r=1}^{50}(7 \times 2^r) = \sum_{r=51}^{100}(7 \times 2^r)$$

Examiner Tips and Tricks

Be careful when more than one letter appears in a sigma notation question.

k IS A VARIABLE BEING USED INSTEAD OF A SPECIFIC NUMBER TO DEFINE THE UPPER LIMIT. CHANGING THE VALUE OF k CHANGES THE VALUE OF THE SUM!

$$\sum_{r=1}^{k}(3r-2) = 1 + 4 + 7 + \ldots + (3k-2)$$

r IS BEING USED HERE AS USUAL TO GIVE THE LIMITS AND DEFINE THE FORMULA FOR THE EXPRESSION

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Worked Example

a) $\sum_{r=50}^{99} (2r+1)$

b) $\sum_{r=k-4}^{12} (3 \times 4^r)$ when $k = 7$

"EVALUATE" HERE MEANS "FIND THE VALUE OF"

a) THIS IS THE SUM $101 + 103 + 105 + \ldots + 199$
ARITHMETIC SERIES WITH $a=101$, $d=2$, AND LAST TERM $l=199$

$S_{50} = \frac{50}{2}(101 + 199) = 7500$

$S_n = \frac{n}{2}(a+l)$

BE CAREFUL HERE! THERE ARE 50 NUMBERS (NOT 49) FROM $r=50$ TO $r=99$. SO $n=50$

b) FOR $k=7$ THIS IS

$$\sum_{r=3}^{12}(3\times4^r) = 3\times4^3 + 3\times4^4 + 3\times4^5 + \ldots + 3\times4^{12}$$
$$= 3\times4^3 + 4(3\times4^3) + 4^2(3\times4^3) + \ldots + 4^9(3\times4^3)$$
$$= 192 + 4(192) + 4^2(192) + \ldots + 4^9(192)$$

THAT IS THE FIRST 10 TERMS OF THE GEOMETRIC SERIES WITH $a=192$ AND $r=4$

$S_{10} = \frac{192(4^{10}-1)}{4-1} = 67\,108\,800$

$S_n = \frac{a(r^n-1)}{r-1}$

BE CAREFUL! THIS r IS THE COMMON RATIO OF THE GEOMETRIC SERIES. IT IS DIFFERENT FROM THE r IN THE SIGMA NOTATION.

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